ONLINE CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025 - 2026)

Enthusiast COURSE

PHASE : ALL PHASE

TARGET : Pre-Medical 2026

Test Type :**MAJOR** Test Pattern :**NEET (UG)**

TEST DATE : 19-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	1	1	4	4	3	2	2	4	3	2	2	3	1	3	4	3	3	2	3	3	4	2	2	3	4	1	2	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	2	3	3	1	4	3	3	2	1	1	4	2	2	2	3	2	2	2	2	2	2	2	1	4	3	2	1	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	3	2	2	1	4	4	3	3	2	1	2	3	3	4	2	2	2	3	2	2	3	3	3	2	4	1	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	1	4	2	4	4	4	1	1	2	1	2	4	4	3	1	3	4	1	2	1	2	2	1	2	1	2	3	3	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	4	3	3	4	4	3	2	1	4	3	2	1	3	1	2	1	4	4	1	1	2	2	2	4	4	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	1	1	1	4	3	4	2	3	1	1	4	2	2	1	1	4	2	1	4	1	4	1	2	4	4	1	3	4

HINT – SHEET

1. Ans (4)

Using the method of dimensions

$[E] \propto [P]^a [A]^b [T]^c$

$\Rightarrow [ML^2T^{-2}] = [MLT^{-1}]^a [L^2]^b [T]^c$

$\Rightarrow a = 1, a + 2b = 2, c - a = -2$

Solving $a = 1, b = 1/2, c = -1$

2. Ans (1)

Work done = Area under the graph from x = 0 to x = 4

$W = 5 \times 2 + \frac{1}{2} \times 2(5 + 10)$

$W = 10 + 15 = 25$ J

3. Ans (1)

$a = \frac{5/3 - 5}{3} = \frac{-10}{9} m/s^2$

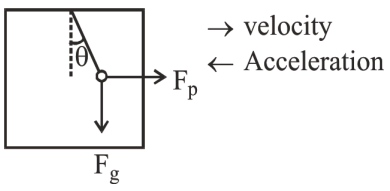
$T\cos\theta = F_g = mg$ ..... (1)

$T\sin\theta = Fp = ma$ ..... (2)

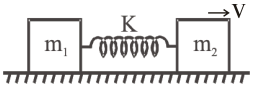
$\frac{(2)}{(1)} = \frac{T\sin\theta}{T\cos\theta} = \frac{|F_P|}{|F_g|}$

$\tan\theta = \frac{F_p}{F_g} = \frac{|ma|}{|mg|} = \frac{1}{9}$

$\theta = \tan^{-1}\left[\frac{1}{9}\right]$



4. Ans (4)



Conservation of linear momentum,

$$M_2V = M_1V' + M_2V' \Rightarrow V' = \frac{M_2V}{M_1 + M_2}$$

Conservation of mechanical energy

$$\frac{1}{2}M_2V^2 = \frac{1}{2}M_1(V')^2 + \frac{1}{2}M_2(V')^2 + \frac{1}{2}Kx^2$$

$$M_2V^2=(M_1+M_2)(V')^2 + Kx^2$$

$$M_2V^2=(M_1+M_2) \times \left(\frac{M_2V}{M_1 + M_2}\right)^2 + Kx^2$$

$$M_2V^2 - \frac{M_2^2V^2}{(M_1 + M_2)} = Kx^2$$

$$x = V\sqrt{\frac{m_1m_2}{(m_1 + m_2)K}}$$

OR

$$\frac{1}{2}kx^2 = \frac{1}{2}\left(\frac{m_1m_2}{m_1 + m_2}\right)v^2$$

$$x = v\sqrt{\frac{m_1m_2}{k(m_1 + m_2)}}$$

5. Ans (4)

Weight of the body = Weigth of the liquid displaced

In water $V\rho_{block}g = \left(\frac{4}{5}V\right)\rho_{water}g$

$$\rho_{block} = \frac{4}{5}\rho_{water} \quad ...(i)$$

In liquid, $V\rho_{block}g = V\rho_{liquid}g$

$$\rho_{block} = \rho_{liquid} \quad ..(ii)$$

From equation (i) and (ii), we get

$$\rho_{liquid} = \frac{4}{5}\rho_{water} = \frac{4}{5} \times 10^3 \text{ kg m}^{-3}$$

$$= 800 \text{ kg m}^{-3}.$$

6. Ans (3)

In normal capillary rise, there is only one meniscus supporting water.

Supported height = h

After taking tube out and opening lower end water column has two menisci



- One at top
- One at bottom

Both provide upward surface tension support

so, total supported length becomes h + h = 2h

Hence length of water column remaining in the tube will be 2h.

7. Ans (2)

$$E = \frac{1}{2}I\omega^2$$

$$I = \frac{2E}{\omega^2}$$

8. Ans (2)

$W_B = T_1 = 0.25 \times 100 = 25$ N

9. Ans (4)

Precision for a set of measurement is seen by least count of instrument.

10. Ans (3)

Here, $K_1 = K_2$, $l_1 = l_2 = 1$m

$A_1 = 2A$, $A_2 = A$

$T_1 = 100°C$ and $T_2 = 70°C$

$\therefore$ Temperature at C be T, then

$$\frac{\Delta Q}{\Delta t} = \frac{K\,2A(100 - T)}{1} = \frac{KA(T - 70)}{1}$$

or $T = 90°C$

11. Ans (2)

FLOT $Q = W + \Delta U$

$Q = +110$ J

$\Delta U = +40$ J

$110 = 40 + W$

$W = +70$ J

12. Ans (2)

$\frac{1}{2} K (A^2 - x^2) = \frac{1}{3}\left(\frac{1}{2}kx^2\right)$

$\Rightarrow A^2 - x^2 = \frac{x^2}{3}$

$\Rightarrow A^2 = \frac{4x^2}{3}$

$\Rightarrow x = \frac{\sqrt{3}}{2} A = 0.87\ A$

$\therefore \frac{x}{A} \times 100\% = 87$

$\Rightarrow x = 87$

13. Ans (3)

As per kepler's law.

Areal speed is constant.

If shaded areas are equal then $t_1 = t_2$

14. Ans (1)

$y_1 = a\sin(\omega t + kx + 0.57)$

$\therefore$ Phase, $\phi_1 = (\omega t + kx + 0.57)$

$y_2 = a\cos(\omega t + kx) = a\sin(\omega t + kx + \frac{\pi}{2})$

$\therefore$ Phase, $\phi_2 = \omega t + kx + \frac{\pi}{2}$

Phase difference, $\Delta\phi = \phi_2 - \phi_1$

$= (\omega t + kx + \frac{\pi}{2}) - (\omega t + kx + 0.57)$

$= \frac{\pi}{2} - 0.57 = 1.57 - 0.57 = 1$ radian

15. Ans (3)

Newton First Law

16. Ans (4)

$\vec{v} = \vec{u} + \vec{a}t = (4\hat{i} + 3\hat{j}) + (0.4\hat{i} + 0.3\hat{j}) \times 10 = 8\hat{i} + 6\hat{j}$

Its magnitude is $|\vec{v}| = \sqrt{(8)^2 + (6)^2} = 10$m/s

17. Ans (3)

COM of :

Semicircular Disc = $\frac{4R}{3\pi}$

Semicircular Ring = $\frac{2R}{\pi}$

Solid hemisphere = $\frac{3R}{8}$

Hemispherical shell = $\frac{R}{2}$

18. Ans (3)

$(2 \times 3)\hat{i} + (1 \times 4)\left(-\hat{i}\right) = (2 + 1)\vec{v}$

$2\hat{i} = 3\vec{v}$

$\vec{v} = \frac{2}{3}\hat{i}$ m/s

19. Ans (2)

$m = 2$kg

$u = 5\ ms^{-1}$

$I = F\Delta t = \Delta p =$ Area under curve

$p_f - p_i = \frac{1}{2} \times 20 \times 10$

$p_f = 100 + 2 \times 5 = 110$ N. sec

20. Ans (3)

$V = \frac{\lambda}{T} \Rightarrow f = \frac{V}{\lambda} = \frac{20}{10} = 2$ m/s

21. Ans (3)

$\varepsilon = \frac{d\phi}{dt} = \frac{d}{dt}[BA\cos 0^\circ] = \frac{d}{dt}\left(B\pi r^2\right)$

$\Rightarrow \varepsilon = B(2\pi r) \cdot \frac{dr}{dt}$

$\Rightarrow \varepsilon = 0.025 \times 2\pi \times 2 \times 10^{-2} \times 10^{-3}$ V olt

$\varepsilon = \pi\mu$ Volt

22. Ans (4)

$qE = qvB$

$v = E/B$

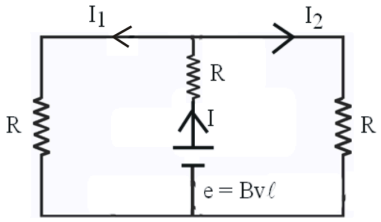
{and $\frac{q}{m} = s$}

$R = \frac{mv}{qB} = \frac{1}{S}\frac{(E/B)}{B} = \frac{E}{B^2S}$

23. Ans (2)

$E = \frac{M|\Delta I|}{\Delta t} = \frac{5 \times (10 - 0)}{5 \times 10^{-4}} = 1 \times 10^5$ volt

24. Ans (2)



$R_{eq} = \frac{R}{2} + R = \frac{3R}{2}$

$I = \frac{2Bv\ell}{3R}$

$I_1 = I_2 = \frac{Bv\ell}{3R}$

25. Ans (3)

$\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$

$\cos\phi = \frac{8}{10} = 0.8$

26. Ans (4)

$V_i = V_0 \sin\omega t$

$I_i = I_0 \sin(\omega t - \frac{\pi}{2}) = -I_0 \cos\omega t$

instantaneous power (P) = $V_i I_i$

$P = -V_0 I_0 \sin\omega t \cos\omega t$

$P = -\frac{V_0 I_0}{2}\sin 2\omega t \quad (\because 2\sin\theta\cos\theta = \sin 2\theta)$

So angular frequency = 2ω

27. Ans (1)

$I = \frac{1}{2}\epsilon_0 \epsilon_r E_0^2 v,\ v = \frac{c}{\sqrt{\mu_r \epsilon_r}}$

28. Ans (2)

$W.D._{\text{ext force}} = -P.E.$

$= -\left(\frac{kq}{a/\sqrt{3}}\right) \times q \times 3$

$= \frac{-3\sqrt{3}kq^2}{a}$

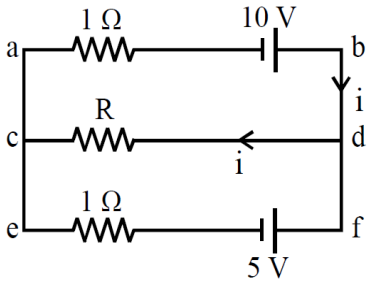
29. Ans (2)

$H = \frac{10^2}{4}$

$H_2 = \frac{10^2}{6} = \frac{4H}{6} = \frac{2H}{3}$

30. Ans (2)

If no current flows through 5V battery, then potential difference across f and e is 5V. So potential difference across d and c is also 5V.



$i = \frac{10}{R+1}$

$iR = 5 \Rightarrow \frac{10}{(R+1)}R = 5 \Rightarrow R = 1\Omega$

31. Ans (2)

Since aperture of lens reduces so brightness will reduce but there will be no effect on size of image.

32. Ans (2)

$r_1 + r_2 = 90$

$r_2 = i_C$

$r_1 = 90 - i_c$

Snell's law on first surface

$\sin i = \mu \sin r_1$

$\sin i = \mu \cos i_c$

$\sin i = \sqrt{\mu^2 - 1}$

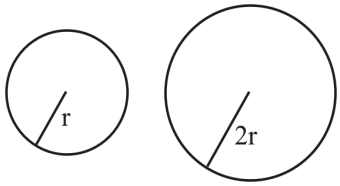
$\mu = \sqrt{1 + \sin^2 i}$

For μ_{max} $\sin^2 i = 1$

So, $\mu_{max} = \sqrt{2}$

33. Ans (3)

After touching & separation of spheres



$\frac{Q}{3}$ $\frac{2Q}{3}$

$\frac{\sigma_1}{\sigma_2} = \frac{\frac{Q/3}{4\pi r^2}}{\frac{2Q/3}{4\pi(2r)^2}} = \frac{2}{1}$

34. Ans (3)

$(I - i_g)\, S = i_g G$

$S = \dfrac{i_g G}{I - i_g} = \dfrac{20 \times 10^{-3}}{1 - 10^{-3}} = 0.02\ \Omega$

35. Ans (1)

For – ve half cycle diode R.B. So NO current how in circuit voltage across resistance is zero.

For +ve half mole diode is F.B. current flow we obtained output across load resistan resistance.

36. Ans (4)

$a \sin\theta = \Delta$

$a \sin\theta = n\lambda$

$\sin 45^\circ = \dfrac{1 \times 6 \times 10^{-7}}{a}$ ...(1)

$a \sin\theta = (2n+1)\dfrac{\lambda}{2}$

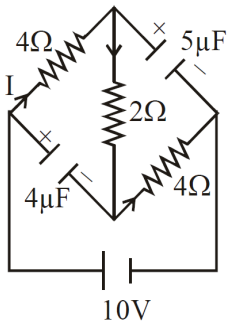
$\sin\theta = \dfrac{5 \times 6 \times 10^{-7}}{a}$ ...(2)

(1)/(2)

$\dfrac{\sin 45^\circ}{\sin\theta} = \dfrac{1}{\left(\frac{5}{2}\right)} = \dfrac{2}{5} \Rightarrow \sin\theta = \dfrac{5}{2} \times \sin 45^\circ$

$\sin\theta = \dfrac{5}{2\sqrt{2}}$ $\boxed{\theta = \sin^{-1}\left(\dfrac{5}{2\sqrt{2}}\right)}$

37. Ans (3)



$I = \dfrac{10}{10} = 1A$

P.d. across 4μF

$-4 \times 1 - 2 \times 1 + v_1 = 0 \Rightarrow v_1 = 6V$

P.d. across 5 μF

$-2 \times 1 - 4 \times 1 + V_2 = 0 \Rightarrow V_2 = 6V$

Energy stored in 4μF & 5μF

$U_1 = \dfrac{1}{2}(4 \times 10^{-6})(6)^2,$

$U_2 = \dfrac{1}{2}(5 \times 10^{-6})(6)^2 \Rightarrow \dfrac{U_2}{U_1} = \dfrac{5}{4} = 1.25$

38. Ans (3)

$n = \dfrac{E\lambda}{hc} = \dfrac{1 \times 10^{-7} \times 200 \times 10^{-9}}{6.6 \times 10^{-34} \times 3 \times 10^{8}} = 1 \times 10^{11}$

So no. of $e^- = \dfrac{1 \times 10^{11}}{10^3} = 10^8 e^-$

$V = \dfrac{q}{4\pi\varepsilon_0 r}$

$= \dfrac{(10^8 \times 1.6 \times 10^{-19}) \times 9 \times 10^9}{4.8 \times 10^{-2}} = 3V$

39. Ans (2)

$\left(\dfrac{\lambda_1}{\lambda_2}\right)^2 = \left(\dfrac{\sqrt{K_2}}{\sqrt{K_1}}\right)^2$

KE in region 1 $(0 \leqslant x \leqslant 1)$

$U_1 = E_0$

TE = KE + PE

$2E_0 = K_1 + E_0$

$K_1 = E_0$

KE in region 2 $(x > 1)$

$U_2 = 0$

TE = KE + PE

$2E_0 = K_2 + 0 \Rightarrow K_2 = 2E_0$

$\left(\dfrac{\lambda_1}{\lambda_2}\right)^2 = \dfrac{K_2}{K_1} = \dfrac{2E_0}{E_2} = 2$

40. Ans (1)

$I_1 = \dfrac{I_0}{2}$

$I_2 = \dfrac{I_0}{2}\cos^2\theta$

$I_3 = \left(\dfrac{I_0}{2}\cos^2\theta\right)\cos^2(90 - \theta)$

$= \dfrac{I_0}{2}\sin^2\theta\cos^2\theta$

$= \dfrac{I_0}{8}\sin^2 2\theta$

41. Ans (1)

Rest mass of parent nucleus is always greater than the product. So rest mass energy is also higher.

42. Ans (4)

$\dfrac{\rho'}{\rho} = \dfrac{m'}{m}\dfrac{r^3}{r'^3} = \dfrac{8}{27}\left(\left(\dfrac{27}{8}\right)^{1/3}\right)^3 = 1$

$\rho' = \rho$

43. Ans (2)

$$R_F = \frac{\Delta V}{\Delta I} = \frac{0.1}{400 \times 10^{-3}} = 0.25\Omega$$

44. Ans (2)

Question Explanation :

Explain the formation of the depletion layer at the p-n junction in a semiconductor.

Concept :

This is based on the formation of the p-n junction.

Explanation:

At the junction, there's a high concentration of holes in the p-type region and electrons in the n-type region.

And the holes diffuses from the p-side to the n-side, and electrons diffuse from the n-side to the p-side leaving behind immobile ionized impurities

Thus a layer is formed called a diffusion layer at the junction.

45. Ans (2)

Energy stored in the capacitor

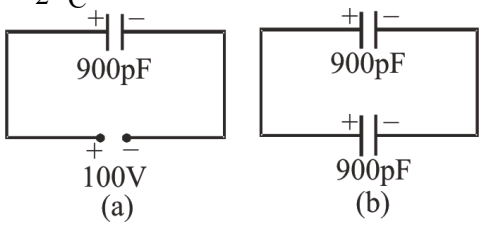
$$= \frac{1}{2}\frac{Q^2}{C}$$

$Q = CV = 900 \times 10^{-12}\ F \times 100V$

$\therefore\ Q = 9 \times 10^{-8} C$

Energy of the capacitor when fully charged

$$= \frac{1}{2}\frac{Q^2}{C} = 4.5 \times 10^{-6} J.$$



The total charge is conserved. In figure (b), total capacitance = C' = 2 × C = 2 × 900 pF.

$$\therefore \text{ Final energy} = \frac{1}{2}\frac{Q^2}{C'} = \frac{1}{2}.\frac{Q^2}{2C}$$

$\therefore$ Final energy

$$= \frac{4.5 \times 10^{-6} J}{2} = 2.25 \times 10^{-6} J.$$

46. Ans (3)

$\Delta H_{eg} \Rightarrow Cl > F > Br > I$

$\Rightarrow S > Se > Te > Po > O$

47. Ans (2)

$NaCl + \text{Conc. } H_2SO_4 + K_2Cr_2O_7 \longrightarrow CrO_2Cl_2$

chromyl chloride

48. Ans (2)

During preparation of Aluminium sulphate, can undergo hydrolysis in water :

$Al_2(SO_4)_3 + 6H_2O \longrightarrow Al(OH)_3 + 3H_2SO_4$

Formation of $Al(OH)_3$ disturbs crystal formation. A small amount of dilute H_2SO_4 is added to maintain acidic medium and suppress hydrolysis.

49. Ans (2)

(A) Pyrosulphuric acid $\longrightarrow$ (r) $H_2S_2O_7$

(B) Peroxy disulphuric acid $\longrightarrow$ (p) $H_2S_2O_8$

(C) Peroxy monosulphuric acid $\longrightarrow$ (q) H_2SO_5

(D) Sulphurous acid $\longrightarrow$ (s) H_2SO_3

50. Ans (2)

IUPAC name of complex $[Pt(NH_3)_4]\ [CuCl_4]$:-

Tetra ammine platinum (II) tetra chlorido Cuprate (II)

51. Ans (2)

$$K = \frac{2.303}{10}\log\frac{0.8}{0.2} = \frac{2.303}{10}\log 4$$

$$\text{and } K = \frac{2.303}{t}\log\frac{0.6}{0.15} = \frac{2.303}{t}\log 4$$

$\therefore$ t = 10 min.

52. Ans (2)

$$m = \frac{1000 \times M}{(1000 \times d) - (M \times M_B)}$$

Where: m = molality of the solution (mol/kg)

M = molarity of the solution (mol/L)

d = density of the solution (g/mL or g/cm^3)

M_B = molar mass of the solute (g/mol)

53. Ans (2)

$$E_{cell} = E^o_{cell} - \frac{0.06}{2}\log\frac{[Mn^{+2}]}{[Sn^{+2}]}$$

$$= (1.18 - 0.14) - 0.03 \log\frac{0.4}{0.04}$$

$= 1.04 - 0.03 \log 10 = 1.01$ volt

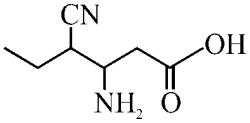
$\Delta G = -\ n\ F\ E_{cell}$

$= -\ 2 \times 96500 \times 1.01$

$= -\ 194.93$ KJ

54. Ans (1)

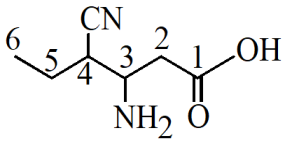
B. Given Data:



C. Concept:

- **IUPAC Nomenclature for Organic Compounds with Multiple Functional Groups:** When a compound has more than one functional group, a priority order determines the parent name and the suffixes/prefixes used.
- **Priority Order:** Carboxylic acids (-COOH) have higher priority than nitriles (-CN) and amines ($-NH_2$).
- **Nitrile as a Substituent:** When a nitrile is not the principal functional group, it is named as a "cyano" substituent.
- **Amine as a Substituent:** When an amine is not the principal functional group, it is named as an "amino" substituent.

D. Answer Explanation:



1. Identify the Main Functional Group: The compound contains a carboxylic acid (-COOH), a nitrile (-CN), and an amine ($-NH_2$). Carboxylic acids have the highest priority, so the parent name will be based on the carboxylic acid.

2. Number the Carbon Chain: Start numbering from the carboxylic acid carbon (which gets the number 1). The chain is six carbons long, making the parent chain "hexanoic acid."

3. Identify and Name Substituents:

- At carbon 3, there is an amino group ($-NH_2$), making it "3-amino."
- At carbon 4, there is a cyano group (-CN), making it "4-cyano."

4. Combine the Parts: Combine the substituent names alphabetically with the parent chain name: "3-amino-4-cyanohexanoic acid."

The correct IUPAC name is **3-amino-4-cyanohexanoic acid**

E. Final Answer: (Option 1).

55. Ans (4)

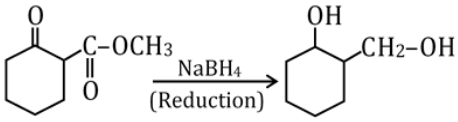
In SN_2 reaction, no intermediate product formed.

56. Ans (3)

Test for sulphur

57. Ans (2)

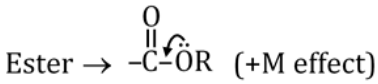
Statement (I) :



It is false statement because $NaBH_4$ cannot reduces ester.

Statement (II) : Ester is more reactive than aldehyde towards nucleophile.

It is a false statement because ester is less reactive than aldehyde to wards nucleophile due to +M effect of –OR group



which reduced the electrophilicity of  group.

∴ Both Statement-I and Statement-II are false.

58. Ans (1)

$F^{\ominus}$ is the best nucleophile in the polar aprotic solvent like DMF.

60. Ans (4)

$$\text{Heat of hydrogenation} \propto \frac{1}{\text{Stability of Alkene}}$$

61. Ans (3)

$2NH_3 \longrightarrow N_2 + 3H_2$

10L 5L 15L

Change in volume = 20 – 10 = 10L

62. Ans (2)

Greater the value of $(n + \ell)$ more will be the energy of orbital. If two orbitals have same $(n + \ell)$ value then the orbital having greater n value have greater energy.

64. Ans (2)

$S - S^- \longrightarrow$ Exothermic process

$O^- \longrightarrow O^{-2}$ Endothermic process

$Sr \longrightarrow Sr^{+2}$ Endothermic process

(inert gas configuration)

$N^- \longrightarrow N$ (Half filled configuration)

65. Ans (2)

$H_2O > H_2S > H_2Se > H_2Te$

Size of Central atom $\uparrow \Rightarrow$ Bond angle $\downarrow$

66. Ans (1)

	C_2^+	$\rightarrow$	C_2
B.O =	1.5	$\rightarrow$	2
	Paramagnetic	$\rightarrow$	diamagnetic

67. Ans (4)

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

(A) (B) (C)

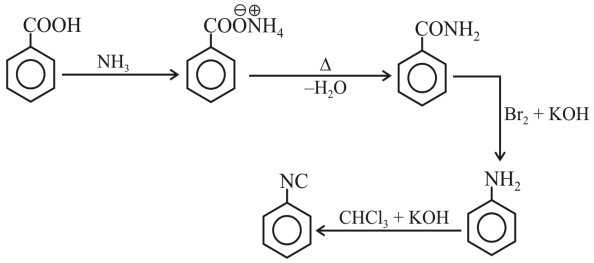
$CaO + H_2O \longrightarrow Ca(OH)_2$

(B) (D)

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 \downarrow$

(D) (C) (Milky)

68. Ans (4)



69. Ans (3)

$\frac{1}{5} \times P_A^o + \frac{4}{5} \times P_B^o = 560$ .....(1)

$\frac{1}{4} \times P_A^o + \frac{3}{4} \times P_B^o = 550$ .....(2)

(1) − (2) × $\frac{4}{5}$

$\frac{P_B^o}{5} = 120 \Rightarrow P_B^o = 600$

From eq (1) $P_A^o = 400$

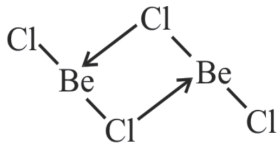
70. Ans (3)

$\Delta_f H = 81.46 - 2 \times 285.78 - (-367.54)$

$= 449 - 571.56$

$= -112.56$ kJ

71. Ans (2)



(Chlorobridged dimer)

72. Ans (1)

$Al^{3+} \longrightarrow$ High Z_{eff}, Smallest size

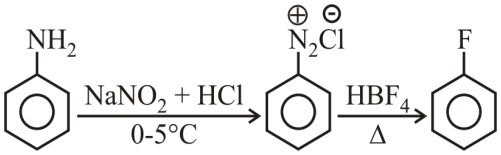
73. Ans (2)

$[Fe(CN)_6]^{-3}$ & $[IrF_6]^{-3} \longrightarrow d^2sp^3$ (inner orbital complex)

74. Ans (3)

Boiling point : Ortho > Para > Meta

75. Ans (3)



76. Ans (4)

Acetone with Ethanol in dry HCl

78. Ans (2)

10^{-8} M HCl $\Rightarrow$ slightly less than 7

10^{-2} M NaOH $\Rightarrow [OH^-] = 10^{-2}$

pOH = 2

pH = 14 − 2 = 12

10^{-2} M $CH_3COOH \Rightarrow CH_3COOH$ is a weak acid so, pH will be less than 2

At 25°C, pH of pure water is 7.

79. Ans (2)

For the reaction having $\Delta n_g = 0$, change in volume of system at equilibrium does not alter number of moles of reactants and products but concentration changes.

80. Ans (3)

$Mn_3O_4 \longrightarrow MnO_4^{2-}$

$+\frac{8}{3}$ (+6)

vf of $Mn_3O_4 = \left(6 - \frac{8}{3}\right) \times 3 = 10$

Q = mole × vf = 0.2 × 10 = 2 F

81. Ans (2)

$[A]_t = [A]_0 - kt$

$0 = [A]_0 - kt$

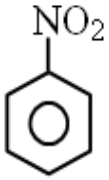
$t = \frac{[A]_0}{k} = \frac{a}{k}$

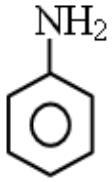
85. Ans (3)

$CH_3-CH_2-OH \xrightarrow{PBr_3} \underset{(A)}{CH_3-CH_2-Br} \xrightarrow[-HX]{Alc.\ KOH} \underset{(B)}{CH_2=CH_2} \xrightarrow{Br_2} \underset{(C)}{\overset{Br}{\overset{|}{CH_2}}-\overset{Br}{\overset{|}{CH_2}}}$

86. Ans (2)

X ⟶ 

Y ⟶ 

M & N ⟶ 

88. Ans (1)

$MnO_4^- + H_2C_2O_4 \longrightarrow Mn^{2+} + CO_2$

v.f. 5 2

$m_{eq.}$ of $KMnO_4 = m_{eq.}$ of $H_2C_2O_4$

$M_1 \times V_1 \times V_f = M_2 \times V_2 \times V_f$

$0.25 \times 20 \times 5 = 0.1 \times V_2 \times 2$

$V_2 = \frac{25}{2} = 12.5$ mL

89. Ans (3)

$\pi_1 = \pi_2$

$i_1C_1RT = i_2C_2RT$

$1 \times \frac{x}{60 \times 100} \times 1000 = 2 \times \frac{0.585 \times 1000}{58.5 \times 100}$

$x = 1.2$

90. Ans (2)

(i) and (iii) are correct.

As_2S_3 is negatively and $Fe(OH)_3$ is positively charged solution.

Lyophilic sols are highly solvated.

91. Ans (2)

Question Explanation :The question asks to identify the false statement(s) about columnar epithelium.

Concept :Structural Organisation in Animals – Epithelial Tissue

Solution

(A) It is made up of tall and slender cells ⟶ correct

(B) Free surface may have microvilli ⟶ correct, especially in absorptive epithelium of intestine

(C) Found in stomach and intestine and helps in secretion and absorption ⟶ correct

(D) Ciliated epithelium is present in bronchioles and fallopian tubes ⟶ correct

(E) They have apical nuclei ⟶ incorrect because nuclei in columnar epithelium are generally located near the basal region, not apical region.

Thus, only statement E is false.

Final Answer :Option 2

92. Ans (1)

NCERT Pg. # 112

93. Ans (4)

Question Explanation:

The question asks the function of the tympanum in frog.

Concept:

Structural Organisation in Animals – Frog Morphology

Solution:

The tympanum in frog is a circular membrane present behind the eye. It acts as an eardrum and receives sound vibrations. Therefore, it helps in hearing.

Final Answer: Option (4)

94. Ans (2)

Question Explanation

The question asks in which phylum true body segmentation is first observed during animal evolution.

Concept

Animal Kingdom – Annelida

Solution

Metameric segmentation means the body is
- **divided into repeated segments.**
- **This feature first appears in Annelida.**

Aschelminthes are unsegmented, while Arthropoda also show segmentation but
- **evolved later.**
- **Mollusca do not show true segmentation.**

Final Answer :Option 2

95. Ans (4)

Question Explanation : The question asks to identify the incorrect statements about different fishes.

Concept : Animal Kingdom – Pisces **Solution**

(A) Poison sting in "Torpedo" ⟶ incorrect

Torpedo is an electric ray having electric
- organs, not poison sting.

(B) Electric organ is found in "Trygon" ⟶ incorrect

Trygon (sting ray) possesses a poison sting,
- not electric organs.

(C) Claspers present in "Scoliodon" ⟶ correct

Male Scoliodon (dogfish) has claspers for
- copulation.

(D) Air bladder present in "Pristis" ⟶ incorrect

Pristis is a cartilaginous fish, and air bladder
- is absent in cartilaginous fishes.

Thus, incorrect statements are A, B and D.

Final Answer : Option 4

96. Ans (4)

Question Explanation : The question asks the function of comb plates in Pleurobrachia.

Concept : Animal Kingdom – Ctenophora

Solution

Pleurobrachia belongs to phylum
- Ctenophora.

Ctenophores possess eight rows of ciliated
- comb plates.

These comb plates beat continuously and
- help the animal move in water.
- Therefore, their main function is locomotion.

Final Answer : Option 4

97. Ans (4)

NCERT, Pg # 47, 49

98. Ans (1)

Question Explanation : The question asks which fish possesses electric organs.

Concept : Animal Kingdom – Pisces

Solution

Torpedo is an electric ray that possesses electric organs used for defense and
- capturing prey.

Carcharodon (great white shark), Exocoetus (flying fish), and Clarias (catfish) do not
- possess electric organs.

Final Answer : Option 1

100. Ans (2)

Question Explanation : The question asks at which vertebral level the trachea divides into right and left bronchi.

Concept : Breathing and Exchange of Gases – Human Respiratory System

Solution

The trachea bifurcates into right and left primary bronchi at the level of the 5th
- thoracic vertebra (T_5).
- This point is called the carina of the trachea.

Final Answer : Option 2

101. Ans (1)

NCERT Page No. # 201

102. Ans (2)

NCERT, Pg # 271

103. Ans (4)

Question Explanation : The question asks the combined name of the glomerulus and Bowman's capsule.

Concept : Excretory Products and their Elimination – Nephron

Solution

The glomerulus is a tuft of capillaries

- present inside Bowman's capsule. Together, the glomerulus and Bowman's
- capsule form the:
  - Renal corpuscle
  - also called Malpighian body

Thus, both terms are correct.

Final Answer : Option 4

104. Ans (4)

NCERT XI Pg # 321

105. Ans (3)

Question Explanation : The question asks to identify the nephron part that is permeable to water but almost impermeable to electrolytes.

Concept : Excretory System – Nephron Function

Solution

(1) PCT ⟶ Reabsorbs both water and electrolytes; not selectively impermeable to electrolytes Incorrect

(2) DCT ⟶ Selective reabsorption under hormonal control; not mainly characterized by this property Incorrect

(3) Descending limb of loop of Henle ⟶ Highly permeable to water but almost impermeable to electrolytes Correct

(4) Collecting duct ⟶ Permeability to water is hormone-dependent (ADH); not always impermeable to electrolytes Incorrect

Final Answer : Option 3

106. Ans (1)

NCERT-Page No. 316

107. Ans (3)

Question Explanation : The question asks which part of the brain controls the pituitary gland.

Concept : Endocrine System – Hypothalamus–Pituitary Regulation

Solution

(1) Epithalamus ⟶ Involved in pineal gland function; does not control pituitary Incorrect

(2) Thalamus ⟶ Relay center for sensory and motor signals; not controlling pituitary Incorrect

(3) Hypothalamus ⟶ Regulates pituitary gland via releasing and inhibiting hormones Correct

(4) Cerebrum ⟶ Responsible for higher brain functions; not involved in pituitary control Incorrect

Final Answer : Option 3

108. Ans (4)

Question Explanation : The question asks to identify the incorrect match between endocrine organs and their secretions.

Concept: Chemical Coordination and Integration – Endocrine Glands

Solution

(1) Heart ⟶ Atrial natriuretic factor: Correct match; ANF is secreted by atrial walls Correct

(2) Kidney ⟶ Erythropoietin: Correct match; stimulates RBC formation Correct

(3) Gastrointestinal tract ⟶ CCK: Correct match; cholecystokinin is secreted by intestinal mucosa Correct

(4) Ovary ⟶ Oxytocin: Incorrect match; oxytocin is released from posterior pituitary, not ovary Incorrect

Final Answer : Option 4

109. **Ans (1)**

Question Explanation : The question asks which hormone acts on Sertoli cells and stimulates secretion of factors that support spermatogenesis.

Concept: Reproductive System – Hormonal Regulation of Spermatogenesis

Solution

(1) FSH ⟶ Acts on Sertoli cells and stimulates secretion of factors (like ABP) that support spermatogenesis Correct

(2) LH ⟶ Acts on Leydig cells to stimulate testosterone secretion Incorrect

(3) Estrogen ⟶ Female hormone; not involved in Sertoli cell stimulation Incorrect

(4) Testosterone ⟶ Produced by Leydig cells; supports spermatogenesis but does not directly act on Sertoli cells in this way Incorrect

Final Answer : Option 1

110. **Ans (2)**

Question Explanation : The question asks about the correct function/definition of placenta in mammals.

Concept : Human Reproduction – Placenta and Foeto-maternal Exchange

Solution

(1) Foetus is attached to myometrium of uterus ⟶ Incorrect; placenta attaches to endometrium, not myometrium Incorrect

(2) Foetus receives oxygen and nourishment from mother's blood ⟶ Correct; placenta facilitates exchange of gases, nutrients, and wastes Correct

(3) Foetus is covered by fetal membranes ⟶ Incorrect; fetal membranes surround the foetus but placenta is not defined by this function Incorrect

(4) Foetus is attached to mother by spermatic cord ⟶ Incorrect; it is umbilical cord, not spermatic cord Incorrect

Final Answer : Option 2

111. **Ans (1)**

NCERT Pg # 46

112. **Ans (2)**

Question Explanation : The question asks to evaluate the correctness of assertion and reason related to the oral contraceptive Saheli.

Concept : Reproductive Health – Oral Contraceptive Pills

Solution

Assertion ⟶ Saheli is a "once a week" oral contraceptive pill with very few side effects and high contraceptive value. Correct

Reason ⟶ Saheli is a new oral contraceptive for females and contains a non-steroidal preparation (centchroman/ormeloxifene). Correct

Link ⟶ The reason explains why Saheli has fewer side effects (being non-steroidal), but it does not explain the "once a week" dosing schedule or high contraceptive value.

Final Answer : Option 2

113. **Ans (2)**

Question Explanation : The question asks the approximate length of each Fallopian tube in the human female reproductive system.

Concept : Human Reproduction – Female Reproductive System (Fallopian Tube)

Solution: Fallopian tube (oviduct) is about 10–12 cm long Correct

Final Answer : Option 2

114. Ans (1)

Question Explanation : The question asks to identify which Assisted Reproductive Technologies (ART) involve In Vitro Fertilisation (IVF).

Concept : Human Reproduction – Assisted Reproductive Technologies (ART)

Solution

(a) ZIFT ⟶ Zygote is formed in vitro (outside body); involves IVF Correct

(b) IUT ⟶ Embryo is transferred to uterus; fertilisation already done in vitro Correct

(c) ICSI ⟶ Sperm is injected into ovum outside body; involves IVF Correct

(d) AI ⟶ Artificial insemination; fertilisation occurs inside body Incorrect

(e) GIFT ⟶ Gametes are transferred to fallopian tube; fertilisation occurs inside body Incorrect

(f) IUI ⟶ Semen placed in uterus; fertilisation occurs inside body Incorrect

Final Answer : Option 1

115. Ans (2)

Question Explanation : The question asks to identify the terminal (permanent) method of contraception advised for the male partner.

Concept : Reproductive Health – Contraceptive Methods

Solution:

(1) Repeated use of condoms ⟶ Temporary barrier method, not permanent Incorrect

(2) Vasectomy ⟶ Surgical permanent (terminal) method for males involving cutting and ligating vas deferens Correct

(3) Coitus interruptus ⟶ Natural method (withdrawal), not permanent Incorrect

(4) Impotency ⟶ Not a contraceptive method; a medical condition Incorrect

Final Answer : Option 2

116. Ans (1)

NCERT XII Page No. # 118, 119, 124

117. Ans (2)

NCERT Page No. # 151, 152

118. Ans (3)

Question Explanation : The question asks to identify the disease that is not transmitted by flies.

Concept : Human Health and Disease – Disease Transmission

Solution

(1) Dysentery ⟶ Can be spread by houseflies carrying pathogens from contaminated food/material Correct

(2) Amoebiasis ⟶ Spread through contaminated food and water; flies can act as mechanical carriers Correct

(3) Filariasis ⟶ Transmitted by female Culex mosquito, not by flies Correct

(4) Amoebic ⟶ Refers to amoebic infection (amoebiasis), which may spread through contaminated food and flies Incorrect

Final Answer : Option 3

120. Ans (1)

Question Explanation : The question asks to evaluate the correctness of two statements related to immunity and lymphoid organs.

Concept : Human Health and Disease – Immunity and Lymphoid Organs

Solution

Statement-I ⟶ Active immunity takes time to develop because the body needs to produce antibodies after exposure to antigen; therefore it is not fast , Incorrect

Statement-II ⟶ Primary lymphoid organs are bone marrow and thymus, not thyroid Incorrect

Final Answer : Option 1

121. Ans (3)

NCERT XII, Pg.# 129, para 7.3

123. Ans (2)

Question Explanation : The question asks which gland secretes hormones during stress and emergency situations.

Concept : Chemical Coordination and Integration – Adrenal Gland

Solution

(1) Thymus gland ⟶ Involved in development of immunity and T-lymphocytes Incorrect

(2) Adrenal gland ⟶ Secretes adrenaline and noradrenaline during stress and emergency ("fight or flight" response) Correct

(3) Pituitary gland ⟶ Master gland controlling other endocrine glands; not mainly responsible for emergency hormones Incorrect

(4) Thyroid gland ⟶ Regulates metabolism through thyroxine secretion Incorrect

Final Answer :Option 2

124. Ans (4)

NCERT-XII, Pg. No. # 230

125. Ans (3)

Question Explanation:

The question asks the incorrect statement related to population ecology.

Concept:

Population Attributes.

Solution:

Birth rate and death rate are population characteristics, not individual characteristics.

Final Answer: Option 3 — Individual has birth rate and death rate.

126. Ans (3)

NCERT XII Pg. No :- 263

127. Ans (4)

NCERT-XII, Pg. # 267

128. Ans (4)

NCERT XII Pg#242-243

129. Ans (3)

NCERT-XII Pg. # 243

133. Ans (3)

NCERT-XII, Pg # 234,235,237(E), 257,258,260(H)

134. Ans (2)

NCERT XII Pg. # 259

135. Ans (1)

NCERT Page No. # 164, 169

136. Ans (3)

NCERT Page No. # 168, 169

137. Ans (1)

NCERT Page No. # 139

138. Ans (2)

Question Explanation:

The question asks the correct relation between two statements about food chain.

Concept: Ecosystem – Energy Flow **Solution:** Statement-I is correct because energy flow in a food chain is unidirectional.

Statement-II is incorrect because shorter food chains provide more available energy, not lesser.

Final Answer: Option (2) Statement I is correct but Statement II is incorrect

139. Ans (1)

Question Explanation:

The question asks the correct relation between assertion and reason.

Concept:

Principles of Inheritance and Variation – Dihybrid Cross

Solution:

Assertion is true because Mendel obtained 9 : 3 : 3 : 1 ratio in dihybrid cross.

Reason is also true because independent assortment states that one pair of traits segregates independently of the other pair.

However, the 9 : 3 : 3 : 1 ratio also depends on the law of dominance, so the reason is not the complete explanation.

Final Answer: Option (4) Both assertion and reason are true but reason is not the correct explanation of assertion.

140. Ans (4)

Question Explanation:

The question asks the genotypic and phenotypic ratio in F_2 generation showing incomplete dominance.

Concept:

Principles of Inheritance and Variation – Incomplete Dominance

Solution:

In incomplete dominance, heterozygote shows intermediate phenotype.

Therefore, both genotypic and phenotypic ratios in F_2 generation are 1 : 2 : 1.

Final Answer: Option (4) Similar phenotypic & genotypic ratio of 1 : 2 : 1

141. Ans (4)

Question Explanation:

The question asks the strand and direction in which Okazaki fragments are synthesized.

Concept:

Molecular Basis of Inheritance – DNA Replication

Solution:

Okazaki fragments are formed discontinuously on the lagging strand in a direction away from the replication fork.

Final Answer: Option (4) Lagging strand, away from fork

142. Ans (1)

Question Explanation:

The question asks the correct set of statements related to genes and RNA processing.

Concept:

Molecular Basis of Inheritance – Gene Expression

Solution:

- (A) Correct — Gene is the functional unit of inheritance.
- (B) Correct — DNA sequence coding for tRNA or rRNA also defines a gene.
- (C) Correct — Exons appear in mature or processed RNA.
- (D) Incorrect — Promoter and regulatory sequences affect inheritance and expression of characters.

Final Answer: Option (1) A, B and C

143. Ans (1)

Question Explanation:

The question asks to fill the blanks related to George Gamow and genetic code.

Concept:

Molecular Basis of Inheritance – Genetic Code

Solution:

George Gamow was a physicist.

He proposed that 4 bases must combine in triplets to code for 20 amino acids.

Final Answer: Option (1) Physicist ; 20

144. Ans (2)

Question Explanation:

The question asks the correct relation between the two statements.

Concept:

Biotechnology and Its Applications – RNAi & Bt Toxin

Solution:

Statement-I is true because RNA interference acts as a cellular defense method in eukaryotes.

Statement-II is false because Bacillus thuringiensis produces proteins toxic to certain insects, not all bacteria.

Final Answer: Option (2) Statement-I is true and statement-II is false

145. Ans (2)

NCERT pg.no.194

146. Ans (2)

Question Explanation:

The question asks the effect of insertion of DNA fragment into pBR322 using Pst-I or Pvu-I.

Concept: Biotechnology – Cloning Vector pBR322

Solution:

Pst-I site is present in the ampicillin resistance gene (amp^R).

Insertion of foreign DNA in this site causes inactivation of amp^R gene.

Therefore, recombinants cannot grow in presence of ampicillin.

Final Answer: Option (2) a & c are correct

147. **Ans (4)**

Question Explanation

The question asks the correct statement about observing DNA fragments after agarose gel electrophoresis.

Concept:

Biotechnology – Gel Electrophoresis

Solution:

DNA fragments are stained with ethidium bromide and observed under UV light.

Final Answer:

Option (4) Ethidium bromide stained DNA can be seen under exposure of UV light.

148. **Ans (4)**

NCERT XIIth Pg # 121, 122

149. **Ans (1)**

Question Explanation:

The question asks the incorrect statement regarding Human Genome Project.

Concept:

Molecular Basis of Inheritance – Human Genome Project

Solution:

The complete sequence of chromosome 1 was published in 2006, not in 2003.

Final Answer:

Option (1) The sequence of chromosome 1 was completed in 2003

150. **Ans (2)**

Question Explanation:

The question asks the scientist who stated that in double stranded DNA, A = T and G = C.

Concept:

Molecular Basis of Inheritance – Chargaff's Rule

Solution:

Erwin Chargaff stated that in double stranded DNA:

- Adenine = Thymine
- Guanine = Cytosine

Final Answer:

Option (2) Erwin Chargaff

151. **Ans (3)**

NCERT XII Pg. # 77

152. **Ans (3)**

Question Explanation:

The question asks the vitamin whose amount increases during conversion of milk into curd.

Concept:

Microbes in Human Welfare – Lactic Acid Bacteria

Solution:

Lactic acid bacteria increase the amount of vitamin B12 during curd formation.

Final Answer: Option (3) Vit-B12

153. **Ans (1)**

NCERT XIIth Pg # 100

154. **Ans (1)**

NCERT, Pg # 201

155. **Ans (1)**

Question Explanation :

The question asks the proportion of dwarf plants produced in a cross between heterozygous tall and dwarf plants.

Concept:

Principles of Inheritance and Variation – Monohybrid Cross

Cross:

Solution: Tt × tt Offspring:

- 50% Tt (Tall)
- 50% tt (Dwarf)

Final Answer: Option (1) 50%

156. **Ans (4)**

Question Explanation :

The question asks the correct relation between assertion and reason.

Concept:

Principles of Inheritance and Variation – Multiple Allelism

Solution:

Assertion is false because multiple allelism means one gene has more than two alleles, not multiple genes.

Reason is true because more than two forms of the same gene may exist in a population.

Final Answer:

Option (4) Assertion is false but Reason is true

157. **Ans (3)**

Question Explanation:

The question asks the pest controlled by proteins encoded by cryIAc and cryIIAb genes.

Concept:

Biotechnology and Its Applications – Bt Cotton

Solution:

cryIAc and cryIIAb genes produce Bt toxin effective against cotton bollworms.

Final Answer: Option (3) Cotton Bollworm

158. **Ans (4)**

NCERT-XII, Pg. No. # 39

159. **Ans (2)**

Question Explanation:

The question asks the number of families represented by the given organisms.

Concept: Living World – Taxonomic Categories

Solution:

- Wheat ⟶ Family Poaceae
- Potato ⟶ Family Solanaceae
- Brinjal ⟶ Family Solanaceae
- Mango ⟶ Family Anacardiaceae

Total families = 3 **Final Answer:** Option (2)

160. **Ans (3)**

NCERT-XI Pg. No. 6

161. **Ans (1)**

NCERT-XI, Pg. # 76

162. **Ans (1)**

Question Explanation

The question asks the correct relation between two statements about root system.

Concept:

Morphology of Flowering Plants – Root System

Solution:

In dicots, direct elongation of radicle forms the primary root.

Fibrous roots arise from the base of the stem.

Final Answer:

Option (1) Both Statement I and Statement II are correct

163. **Ans (4)**

Question Explanation

The question asks the correct statements about internal structure of dicot stem.

Concept:

Anatomy of Flowering Plants – Dicot Stem

Solution:

- Hypodermis is collenchymatous.
- Pericycle occurs in patches above phloem.
- Endodermis contains starch grains.

All statements are correct. **Final Answer:** Option (4) All the above

164. **Ans (2)**

Question Explanation:

The question asks the correct statement about T.S. of monocot root.

Concept:

Anatomy of Flowering Plants – Monocot Root

Solution:

Monocot root usually has polyarch condition, i.e., more than six xylem bundles.

Final Answer:

Option (2) It has more than six xylem bundles

165. Ans (2)

Question Explanation:

The question asks which photosynthetic pigment absorbs maximum blue light.

Concept:

Photosynthesis in Higher Plants – Photosynthetic Pigments

Solution:

Chlorophyll a shows maximum absorption in blue-violet region of light.

Final Answer: Option (1) Chlorophyll a

166. Ans (1)

Question Explanation:

The question is based on light reaction of photosynthesis.

Concept:

Photosynthesis in Higher Plants – Photolysis of Water

Solution:

Assertion is true: H^+ ions accumulate in thylakoid lumen.

Reason is true: Water splitting occurs on inner side of membrane.

But Reason does not correctly explain accumulation of protons.

Final Answer:

Option (4) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

167. Ans (1)

Question Explanation

The question is based on respiratory metabolism and common intermediate of biomolecules.

Concept:

Respiration – Amphibolic pathway

Solution:

Carbohydrates are broken into pyruvic acid, fats into acetyl CoA (via β-oxidation), and proteins into various intermediates after deamination, many of which converge into acetyl CoA.

Thus, the common metabolite in breakdown of fats, carbohydrates and proteins is acetyl CoA.

Final Answer: Option (1) Acetyl CoA

168. Ans (4)

Question Explanation:

The question is based on ATP formation during respiration.

Concept:

Respiration – Substrate Level Phosphorylation

Solution:

Substrate level phosphorylation occurs in glycolysis (formation of ATP directly from phosphorylated intermediates) and in Krebs cycle (during conversion of succinyl-CoA to succinate).

Final Answer:

Option (4) Krebs cycle and glycolysis

169. Ans (2)

Question Explanation:

The question is based on plant growth regulators and fruit ripening.

Concept:

Plant Growth Regulators – Ethylene

Solution:

Ethylene is a volatile plant hormone released by ripened fruits like oranges. It accelerates (hastens) the ripening of stored unripe fruits such as bananas.

Final Answer:

Option (2) A-Hastened, B-Ethylene

170. Ans (1)

Question Explanation:

The question is based on structure of anther in flowering plants.

Concept:

Flowering Plants – Structure of Anther

Solution:

A typical angiosperm anther is bilobed, dithecous and tetrasporangiate.

Final Answer:

Option (1) Bilobed, dithecous, tetrasporangiate

171. **Ans (4)**

NCERT-XI Pg. # 21

172. **Ans (1)**

Question Explanation:

The question is based on pigments present in different algae.

Concept:

Algae – Pigments (Chlorophyll a and b)

Solution:

Chlorophyll a and b are present in green algae (Chlorophyceae): Chara, Spirogyra, Volvox.

Ectocarpus has chlorophyll a and c.

Polysiphonia and Porphyra have chlorophyll a and phycobilins.

So, total examples with chlorophyll a and b = 3.

Final Answer: Option 1 — Three

173. **Ans (4)**

Question Explanation:

The question is based on sexual reproduction in flowering plants and seed types.

Concept:

Flowering Plants – Pollination and Seed **Solution:**

- Cleistogamous flowers are always autogamous ⟶ correct
- Self-incompatibility prevents self-pollination (outbreeding device) ⟶ correct
- Black pepper is an example of perispermic seed ⟶ correct
- Pea and groundnut are non-albuminous (exalbuminous), not albuminous ⟶ incorrect statement

Final Answer: Option 4 — Pea & groundnut are example of albuminous seeds

174. **Ans (1)**

Question Explanation:

The question is based on plant growth and development processes.

Concept:

Plant Growth and Development

Solution:

A. Plasticity ⟶ IV (Formation of different structures in plants in response to environment or phases of life)

B. Differentiation ⟶ II (Cells losing protoplasm to form a tracheary element)

C. Development ⟶ I (All changes from germination of seed to senescence)

D. De-differentiation ⟶ III (Formation of cork cambium from parenchyma cells)

Final Answer: Option 1 — A-IV, B-II, C-I, D-III

175. **Ans (2)**

Question Explanation

The question is based on enzyme classification.

Concept

Enzymes – Classification (IUB system)

Solution

Enzymes that catalyse hydrolysis of ester, ether, peptide, glycosidic, C–C, C–halide or P–N bonds are classified as hydrolases.

Final Answer

Option 2 — Hydrolases

176. **Ans (4)**

NCERT XI Page No. # 35, 38, 39

177. **Ans (4)**

Question Explanation

The question is based on plant growth and meristematic activity.

Concept

Plant Growth and Development – Meristems and Growth

Solution

(A) True — Open form growth occurs due to continuous addition of new cells by meristems.

(B) True — Vascular cambium and cork cambium increase girth (secondary growth).

(C) True — Plants show unlimited growth due to presence of meristems throughout life.

(D) True — Meristems have ability to divide and self-perpetuate.

Final Answer

Option 4 — All the statement are correct

178. **Ans (1)**

Question Explanation

The question is based on aerobic respiration and Krebs cycle.

Concept

Respiration – Krebs Cycle (TCA Cycle)

Solution

Krebs cycle is also called TCA cycle because the first stable product formed is citric acid, which is a tricarboxylic acid containing three carboxyl (–COOH) groups. Therefore, the reason correctly explains why it is called the tricarboxylic acid cycle.

Final Answer

Option 1 — Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

179. **Ans (3)**

Question Explanation

The question is based on biomolecules, specifically structure of fats.

Concept

Biomolecules – Lipids (Triglycerides)

Solution

In a typical fat (triglyceride), glycerol is linked with fatty acids through ester bonds formed between –OH group of glycerol and –COOH group of fatty acids.

Final Answer

Option 3 — Ester bond

180. **Ans (4)**

Question Explanation

The question is based on DNA replication enzymes.

Concept

Molecular Basis of Inheritance – DNA Replication

Solution

Statement I is false because DNA-dependent DNA polymerase synthesizes DNA only in the $5' \longrightarrow 3'$ direction, not $3' \longrightarrow 5'$.

Statement II is true because DNA polymerase cannot initiate DNA synthesis on its own; it requires a primer.

Final Answer

Option 4 — Statement-I is false but Statement-II is true.